MA1522 Linear Algebra for Computing
L2 Lesson 6



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Semester 1, 2026/27

Minors, Cofactors and Adjoint of a Matrix

Let $\boldsymbol{A}$ be a square matrix.

1. The (i,j)-**minor** of $\boldsymbol{A}$, denoted $\boldsymbol{M}_{ij} = \boldsymbol{M}_{ij}(\boldsymbol{A})$, is the matrix obtained from $\boldsymbol{A}$ by removing its i-th row and j-th column.
2. The (i,j)-**cofactor** of $\boldsymbol{A}$, denoted A_{ij}, is
$$A_{ij} = (-1)^{i+j} \det(\boldsymbol{M}_{ij}).$$
3. The **adjoint** of $\boldsymbol{A}$, denoted $\mathbf{adj}(\boldsymbol{A})$, has its (i,j)-entry equal to A_{ji}.

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Theorem

Let $\boldsymbol{A} = (a_{ij})_{n\times n}$ *be a square matrix.*

1. $\det(\boldsymbol{A}) = \sum_{j=1}^{n} a_{ij}A_{ij}$ *for all* i *(expand along* i*-th row).*
2. $\det(\boldsymbol{A}) = \sum_{i=1}^{n} a_{ij}A_{ij}$ *for all* j *(expand along* j*-th column).*
3. $\boldsymbol{A}\,\mathbf{adj}(\boldsymbol{A}) = \det(\boldsymbol{A})\boldsymbol{I}_n = \mathbf{adj}(\boldsymbol{A})\,\boldsymbol{A}$.

Corollary

Let $\boldsymbol{A}$ *be a square matrix. Then* $\boldsymbol{A}$ *is invertible if and only if* $\det(\boldsymbol{A}) \neq 0$*, in which case*

$$\boldsymbol{A}^{-1} = \frac{1}{\det(\boldsymbol{A})}\,\mathbf{adj}(\boldsymbol{A}).$$

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Question 1

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Let $\boldsymbol{A} = \begin{pmatrix} -x & 1 & 0 \\ 0 & -x & 1 \\ 2 & -5 & 4-x \end{pmatrix}$.

a. Compute $\det(\boldsymbol{A})$ in terms of x by expanding along
   i. the first row;
   ii. the third column.

b. Find all values of x such that $\boldsymbol{A}$ is a singular matrix.

(Cf. Pg 139 of Chapter 2 Lecture Slides.)

Question 1: Solution

(a) (i)

$$\begin{aligned}\det(\boldsymbol{A}) &= (-x)\det\begin{pmatrix}-x & 1\\ -5 & 4-x\end{pmatrix} + (-1)\det\begin{pmatrix}0 & 1\\ 2 & 4-x\end{pmatrix}\\ &= (-x)(x^2-4x+5) - (-2) = -x^3+4x^2-5x+2.\end{aligned}$$

(ii)

$$\begin{aligned}\det(\boldsymbol{A}) &= (-1)\det\begin{pmatrix}-x & 1\\ 2 & -5\end{pmatrix} + (4-x)\det\begin{pmatrix}-x & 1\\ 0 & -x\end{pmatrix}\\ &= (-1)(5x-2) + (4-x)x^2 = -x^3+4x^2-5x+2.\end{aligned}$$

(b) $\det(\boldsymbol{A}) = -x^3+4x^2-5x+2 = -(x-1)^2(x-2)$. Thus

$$\boldsymbol{A} \text{ is singular} \Leftrightarrow \det(\boldsymbol{A}) = 0 \Leftrightarrow x = 1 \text{ or } x = 2.$$

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Question 2

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Let $\boldsymbol{A} = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 2 & 1 \\ 3 & 0 & 6 \end{pmatrix}$. Compute $\mathbf{adj}(\boldsymbol{A})$ and use it to find $\boldsymbol{A}^{-1}$.

(Cf. Pg 135, 161, 163 of Chapter 2 Lecture Slides.)

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(Cf. Pg 135, 161, 163 of Chapter 2 Lecture Slides.)

Solution.

$$\mathbf{adj}(\boldsymbol{A}) = \begin{pmatrix} 12 & 6 & -5 \\ 3 & 0 & -1 \\ -6 & -3 & 2 \end{pmatrix}.$$

Expand along the first column of $\boldsymbol{A}$:

$\det(\boldsymbol{A}) = 1 \det \begin{pmatrix} 2 & 1 \\ 0 & 6 \end{pmatrix} + 3 \det \begin{pmatrix} -1 & 2 \\ 2 & 1 \end{pmatrix} = 1 \cdot 12 + 3 \cdot (-5) = -3$. So

$$\boldsymbol{A}^{-1} = \frac{1}{\det(\boldsymbol{A})} \mathbf{adj}(\boldsymbol{A}) = \begin{pmatrix} -4 & -2 & \frac{5}{3} \\ -1 & 0 & \frac{1}{3} \\ 2 & 1 & -\frac{2}{3} \end{pmatrix}.$$

□

Properties of determinants

Let $\boldsymbol{A}$ and $\boldsymbol{B}$ be square matrices of order n. Then:

1. $\det(\boldsymbol{A}^T) = \det(\boldsymbol{A})$;
2. $\det(c\boldsymbol{A}) = c^n \det(\boldsymbol{A})$;
3. $\boldsymbol{A}$ is invertible if and only if $\det(\boldsymbol{A}) \neq 0$, in which case

$$\det(\boldsymbol{A}^{-1}) = \frac{1}{\det(\boldsymbol{A})};$$

4. $\det(\boldsymbol{AB}) = \det(\boldsymbol{A}) \det(\boldsymbol{B})$.

Question 3

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Let $\boldsymbol{A}$, $\boldsymbol{B}$ and $\boldsymbol{P}$ be square matrices of the same size, then assume that $\boldsymbol{P}$ is invertible. Which of the following statements is/are true?

a. $\det(-\boldsymbol{A}) = -\det(\boldsymbol{A})$.

b. $\det(\boldsymbol{AB}) = \det(\boldsymbol{BA})$.

c. $\det(\boldsymbol{A} + \boldsymbol{B}) = \det(\boldsymbol{A}) + \det(\boldsymbol{B})$.

d. $\det(\boldsymbol{PA}(\boldsymbol{P}^{-1})^T) = \det(\boldsymbol{A})$.



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b. $\det(\boldsymbol{AB}) = \det(\boldsymbol{BA})$.

$$\det(\boldsymbol{AB}) = \det(\boldsymbol{A})\det(\boldsymbol{B}) = \det(\boldsymbol{B})\det(\boldsymbol{A}) = \det(\boldsymbol{BA}).$$

c. $\det(\boldsymbol{A}+\boldsymbol{B}) = \det(\boldsymbol{A}) + \det(\boldsymbol{B})$.

Let $\boldsymbol{A} = \boldsymbol{I}_2$ and $\boldsymbol{B} = -\boldsymbol{I}_2$. Then

$$\det(\boldsymbol{A}+\boldsymbol{B}) = \det(\boldsymbol{I}_2 + (-\boldsymbol{I}_2)) = \det(\boldsymbol{0}_{2\times 2}) = 0,$$
$$\det(\boldsymbol{A}) + \det(\boldsymbol{B}) = \det(\boldsymbol{I}_2) + \det(-\boldsymbol{I}_2) = 1 + 1 = 2.$$

d. $\det(\boldsymbol{P}\boldsymbol{A}(\boldsymbol{P}^{-1})^T) = \det(\boldsymbol{A})$.

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Question 4

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Let $\boldsymbol{A}$ be an invertible square matrix of order n, and let c be a nonzero real constant. Express $\det(\mathbf{adj}(c\boldsymbol{A}))$ in terms of $\det(\boldsymbol{A})$ and c.

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Let $\boldsymbol{B} = c\boldsymbol{A}$. Then $\det(\boldsymbol{B}) = \det(c\boldsymbol{A}) = c^n \det(\boldsymbol{A}) \neq 0$,

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$$\begin{aligned}\det(\boldsymbol{B})\det(\mathbf{adj}(\boldsymbol{B})) &= \det(\boldsymbol{B}\,\mathbf{adj}(\boldsymbol{B})) = \det(\det(\boldsymbol{B})\boldsymbol{I}_n) \\ &= \det(\boldsymbol{B})^n \det(\boldsymbol{I}_n) = \det(\boldsymbol{B})^n.\end{aligned}$$

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Thus, $\det(\mathbf{adj}(\boldsymbol{B})) = (\det(\boldsymbol{B}))^{n-1}$ since $\det(\boldsymbol{B}) \neq 0$,

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Thus, $\det(\mathbf{adj}(\boldsymbol{B})) = (\det(\boldsymbol{B}))^{n-1}$ since $\det(\boldsymbol{B}) \neq 0$, and so

$$\begin{aligned}\det(\mathbf{adj}(c\boldsymbol{A})) &= \det(\mathbf{adj}(\boldsymbol{B})) = \det(\boldsymbol{B})^{n-1} \\ &= (c^n \det(\boldsymbol{A}))^{n-1} = c^{n(n-1)}(\det(\boldsymbol{A}))^{n-1}.\end{aligned}$$

□

Next Lesson ...

We will explore **Sections 3.1–3.3**.
Please watch the video clips for these sections before then.